ABDOULIE BALISA

Data Scientist

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SUMMARY

Statistics undergraduate and Mastercard Foundation Scholar working in data science, machine learning and statistical modeling. Comfortable in Python, R and SQL, with project experience in predictive modeling, Bayesian inference, time-series forecasting, survival analysis and natural language processing. Three working papers, all asking whether a model or the uncertainty it reports survives being moved to a country it was never fitted on. I build projects end to end, from the data pipeline through the model to a working interface, and I document the validation as carefully as the result.

WORKING PAPERS

Manuscripts, unrefereed and not submitted. Full texts at balisa50.github.io.

Balisa, A. (2026). *Site-invariant representations for cough-audio screening: removing the confound does not recover disease signal.* Adversarial domain-invariance removes the recording site from cough representations, in eight of nine held-out countries ($p = 0.012$), while a null model using country prevalence alone outscores the classifier (0.741 against 0.489). Reported as a negative result.

Balisa, A. (2026). *Do prediction intervals for satellite-based poverty estimates survive a national border? A pre-registered evaluation in West Africa.* Twelve DHS surveys, 6,706 clusters, 155,796 households, with The Gambia withheld from every stage and evaluated once. Conformal coverage transfers on average and disperses 2.8 to 4.7 times more than sampling permits for an individual country.

Balisa, A. (2026). *Reconstructed mortality series understate forecast uncertainty: evidence from The Gambia and thirty countries.* Median innovation standard deviation is 2.95 where civil registration is complete and 1.02 where it is not (Mann–Whitney $p < 0.0001$), which accounts for prediction intervals three to five years wide against the United Nations' twenty.

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, TypeScript, MATLAB
- **Statistics & Modeling:** statistical inference, hypothesis testing, regression analysis, Bayesian inference (PyMC, MCMC), survival analysis, time-series forecasting, conformal prediction, Monte Carlo simulation, stochastic processes
- **Machine Learning:** scikit-learn, PyTorch, classification, clustering, dimensionality reduction, generative models, domain adaptation, feature engineering, model validation
- **Data Analysis:** pandas, NumPy, SciPy, ETL pipelines, data cleaning, exploratory data analysis, A/B testing
- **NLP & AI:** sentence embeddings, semantic search, retrieval-augmented generation (RAG), large language models, prompt design
- **Visualization & BI:** Matplotlib, Seaborn, Plotly, Power BI, Tableau
- **Tools & Deployment:** Git, Jupyter, FastAPI, Docker, PostgreSQL, Next.js, LaTeX

EXPERIENCE

Data Science and Machine Learning Developer — 2024 – Present
Independent — *Remote*

- Built and deployed eleven data science and machine learning projects end to end, each open-sourced and running in production. Detail below; they are not restated here.
- Work spans Bayesian forecasting, generative models for tabular data, credit scoring, survival analysis and domain adaptation, with the validation documented as carefully as the headline number
- Repeatedly found and corrected errors in my own published results, including a projection re-based after the 2024 census and a model whose apparent accuracy was an artefact of the train-test split

Data Automation Assistant — Mar – Jun 2024
Lujo Heights Real Estate — *Remote, Nigeria*

- Built a lead-classification system from CRM engagement data that tagged prospects as hot, warm or cold, replacing manual sorting by the sales team
- Wrote Python ETL workflows and a dashboard for tracking sales conversion

Peer Tutor, Data Science and AI — 2024
TechUp Africa — *Remote*

- Taught Python, statistics and machine learning to student groups, reviewed their code and supported them through project work

Freelance Web Developer 2026
BS Real Estate *The Gambia*

- Delivered a client property website with a private admin dashboard for listing management (Next.js, Prisma, PostgreSQL), end to end

SELECTED PROJECTS

The Gambia 2074: Bayesian Population Projection Python, PyMC, NumPy, pandas

- Projected national population to 2074 for a country with no death registration, using Lee-Carter mortality in classical, Bayesian Poisson and coherent multi-country forms feeding a cohort-component model
- Validated the engine against the United Nations projection to within 1%, and backtested the mortality forecast to a mean error of 0.65 years over 13 held-out years
- Identified that published UN figures sit about 13% above the 2024 census, and re-based the projection accordingly balisa50.github.io/research/gambia-2074

NOVA: Synthetic Financial Data Engine Python, PyTorch, FastAPI, Docker, Next.js

- Implemented a Conditional Tabular GAN from the original paper in PyTorch, including mode-specific normalization, WGAN-GP and conditional training-by-sampling, without using an existing library
- Reported statistical similarity 0.94, correlation L1 0.05, train-on-synthetic test-on-real 0.92 and a distance-to-closest-record privacy ratio of 1.10
- Added a rule-based generation mode for populations with no source dataset nova-fin.vercel.app

Credit Risk Scorecard for West African Microfinance Python, scikit-learn, pandas

- Built a Basel II compliant scorecard with Weight of Evidence and Information Value feature selection, logistic regression and points conversion
- Ran a full validation gauntlet (Gini, KS, ROC, PSI) on a time-based holdout after finding that a random split understated population drift
- Added multi-scenario stress testing for drought, currency crisis and pandemic credit-risk-ab.vercel.app

Additional projects: Life Insurance Risk Model (Gompertz-Makeham mortality, Cox proportional hazards, Monte Carlo value at risk); AYAT (sentence embeddings, browser-side semantic search over 6,236 texts); Gambia Legal Aid (retrieval-augmented generation over statute with citation validation); DalasiPulse (Prophet and SARIMA ensemble). Full write-ups at balisa50.github.io.

EDUCATION

BSc Statistics 2024 – Present
Kwame Nkrumah University of Science and Technology (KNUST) *Kumasi, Ghana*

- Mastercard Foundation Scholar
- Coursework: Probability Theory, Statistical Inference, Regression Analysis, Stochastic Processes, Linear Algebra, Machine Learning, R Programming

CERTIFICATIONS

Associate Data Scientist, DataCamp (2024) • Data Science Career Track, DataCamp (2024) • AI Ethics, DataCamp (2024) • Data Science Bootcamp, Axia Africa (2024) • Software Engineering, PLP Academy (2025) • AI Engineering, Udemy (2026)

LEADERSHIP

Vice President, previously Organising Secretary 2025 – Present
GAMTECH, Gambian Students Association at KNUST *Kumasi, Ghana*
Lead the executive committee, assigning portfolios and holding the team to its delivery dates, and plan the year's events including orientation for newly arrived Gambian students. Previously Organising Secretary, managing meetings and official correspondence.